

DAVID IZUCHUKWU GILBERT

Chemical Engineering Student | Full-Stack Developer

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PROFESSIONAL PROFILE

Chemical Engineering undergraduate at the University of Lagos with a 4.94/5.00 official CGPA through 100 Level and completed projects spanning process analysis, materials and biofiltration, clean-energy design, accessible computing, full-stack software and applied machine learning. Combines engineering reasoning with the ability to design, build and communicate working technical prototypes.

EDUCATION & ACADEMIC RECORD

Bachelor of Engineering (B.Eng.), Chemical Engineering

University of Lagos, Lagos, Nigeria | 2024-2029 (expected) | 200 Level

- Official CGPA: 4.94/5.00 through 100 Level.
- Relevant study includes thermodynamics, material and energy balances, fluid mechanics, heat transfer, engineering materials, mathematics and laboratory work.

Senior Secondary Education

Elite Crop College | 2020-2023

- Overall Best Senior Secondary School Student, 2023.
- Head Boy, SS3, 2023.
- SAT: 1360/1600, 2024. JAMB UTME: 302/400, 2024.
- WAEC, 2023: three A grades across nine subjects. NECO, 2023: five A grades across nine subjects.

ENGINEERING PROJECTS & COMPETITIONS

Cement Production Conversion Gap Analysis

DCP University Engineering Challenge 2026 | Team Kilvora | Team Leader | Top 20 of 300+ submissions (Top 5%)

- Led ANSYS structural and thermal analysis of a cylindrical mill-shell and liner model and carried out supporting engineering calculations, scalability analysis and technical research.
- Contributed to a team analysis estimating a 27.6% conversion gap and modelling a pathway to 85.0% conversion; financial estimates combined researched inputs with team calculations.
- Prepared technical and business-case submission materials with a two-person team. Team profiles were forwarded to Dangote Cement Plc after the Top 20 stage.

TLUD Gasifier Cookstove and PAYG Adoption Model

The Industry Discourse 5.0 Energy Startup Challenge | Team Axel | Team Leader | Top 20 semifinalist | 2026

- Directed a four-person Chemical Engineering team developing a design-stage Top-Lit Updraft gasifier cookstove concept paired with a pay-as-you-go mobile-money adoption model.
- Directed and reviewed fuel, market and financial research; vetted assumptions and revised projections for scalability and commercial accessibility.

Klova Passive Biofiltration Cartridge

Prototype Inter-Departmental Engineering Challenge 1.0 | Team CHG | Team Leader | 4th among 10 UNILAG Engineering departmental teams | 2026

- Originated the Klova concept and led a six-person team through repeated design revisions, laboratory preparation, prototype construction and testing.
- Built a gravity-fed pipe test arrangement and cartridge using wire and nylon mesh, charcoal/biochar, granite and soil; incorporated Bacillus and Pseudomonas following laboratory guidance and autoclaving work.
- Created two Blender 3D process visualisations. Stage 1 score: 90.5/100, the highest Chemical Engineering score. Estimated bill of materials: approximately ₦11,500.
- Video 1 | Video 2

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Engineering & Software Projects

GraphSense Accessible Chart-to-Audio Prototype

AFRETEC-UNILAG MathWorks Category | Team Sync | Winning Team | 2026

- Solely developed the MATLAB implementation and produced the demonstration video for a four-person team project based on Mumeen's original concept; Goodness prepared slides and supporting presentation materials.
- Built a graphical interface that accepts data and converts line graphs, bar charts and scatter plots into changing audio pitch, with strongest performance on line graphs.
- Conducted informal blindfold usability trials with sighted participants and presented the working prototype to judges.

SOFTWARE PROJECTS

Swift | Owner and Full-Stack Developer

Active student-learning platform | 2023-Present | swift-md.vercel.app

- Independently rebuilt and now maintain the platform after its initial collaborative phase.
- Organises UNILAG Chemical Engineering course documents through the CCMAS 300 Level curriculum and provides curated objective, multiple-choice and theory tests.
- Serves more than 20 real students and has received strong user feedback. Uses React, JavaScript, Supabase and supporting Python tooling.

Vouch | Frontend Engineer

Squad Hackathon 3.0 | Ranked 63rd of 1,600 entries | vouchsignal.vercel.app

- Built the trader-facing React interface for a transaction-based trust-scoring system aimed at Nigeria's informal economy.
- Implemented score visualisation, financial-tier progression and live score updates using Server-Sent Events.
- Stack included React, Vite, Tailwind CSS, TypeScript, Node.js, Express, Prisma, Supabase and PostgreSQL.

Resonance | Frontend Engineer

DSN × BCT LLM Agent Challenge | project-resonance.vercel.app

- Developed the frontend for a two-agent system designed to predict how a target user might rate a product and identify promising audience segments.
- Integrated authentication and persistent prediction history using Firebase and Firestore.
- Worked with React, Vite, Tailwind CSS, FastAPI, Python and Groq API in a four-person team.

Neuronode | Solo Developer

BuildQuik Challenge | Full-stack AI learning tool

- Designed and built a DSA tutor that uses guided questioning instead of immediately revealing answers.
- Created prompting logic that identifies where a learner is stuck and selects the next useful question.
- Built with React, Node.js, Express, OpenAI API and Tailwind CSS.

Urban Brand Creatives Website | Solo Full-Stack Developer

Client website project

- Designed and built the complete company website, handling the user interface, application logic and deployment workflow.

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Experience, Skills & Recognition

PROFESSIONAL & TECHNICAL EXPERIENCE

Backend Intern

HNG Internship Cohort 13 | Backend Track | October–December 2025

- Completed Stages 0–3 through time-bound backend-development tasks involving APIs, testing, documentation and version-controlled delivery.

AI/ML Track Participant

ECX UNILAG | 2025–Present

- Built a credit-card fraud-detection model on approximately 568,000 transactions, achieving 99.97% recall on a separate test set.
- Applied feature engineering, SMOTE, Random Forest and XGBoost using Python, pandas and scikit-learn.

AI/ML Fellow

3MTT Cohort 4 | Ongoing

- Participating in the 3 Million Technical Talent AI/ML pathway.

Operator-Training Simulation Exposure

National RTSim OTS Workshop | August 2026

- Gained introductory exposure to RTSim.Career and operator-training concepts.

TECHNICAL SKILLS

Software development	JavaScript, TypeScript, Python, React, Vite, Tailwind CSS, Node.js, Express, FastAPI, REST APIs
Data and platforms	PostgreSQL, Supabase, Firebase, Firestore, Prisma ORM, Git, GitHub, Vercel
Machine learning	pandas, scikit-learn, XGBoost, Random Forest, SMOTE, Jupyter Notebook, Google Colab
Engineering analysis	ANSYS, MATLAB, JMP; mass and energy balances; thermodynamics; process-control fundamentals
Design and visualisation	Working familiarity with AutoCAD, Fusion 360 and Blender
Language	English — Fluent

CERTIFICATIONS

- AI/ML Essentials — ECX 6.0, Engineering Career Expo.
- Generative AI Foundation — UNICEF / Nigeria Learning Passport.
- Generative AI Medium — UNICEF / Nigeria Learning Passport.
- Generative AI Master — UNICEF / Nigeria Learning Passport.

PROFESSIONAL MEMBERSHIPS

- Student Member, American Institute of Chemical Engineers (AIChE).
- Student Member, Nigerian Society of Chemical Engineers (NSChE).

AWARDS & RECOGNITION

- AFRETEC-UNILAG MathWorks Category — Winning Team, Team Sync, 2026.
- DCP University Engineering Challenge — Top 20 of 300+ submissions; Top 5%, Team Kilvora, 2026.
- PIDEC 1.0 — 4th among 10 departmental teams; 90.5/100 and highest Chemical Engineering Stage 1 score, 2026.
- TID 5.0 Energy Startup Challenge — Top 20 semifinalist, Team Axel, 2026.
- Squad Hackathon 3.0 — 63rd of 1,600 entries, Vouch.
- Overall Best Senior Secondary School Student — Elite Crop College, 2023.

LEADERSHIP

- Head Boy — Elite Crop College, 2023.
- Team Leader — Team Kilvora, DCP University Engineering Challenge 2026.
- Team Leader — Team Axel, TID 5.0 Energy Startup Challenge 2026.
- Team Leader — Team CHG, PIDECE 1.0, 2026.